

NT-C-031 | CCNA | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Networking
Target learners	Learners preparing for junior networking and NOC support roles.
Prerequisites	Basic computer and networking awareness.
Approved duration	12-16 weeks / 140-180 guided hours.
Final outcome	Configure and troubleshoot routed and switched networks using secure foundational practices.
Assessment profile	Infrastructure/Security

Approved tools / platforms

Cisco IOS lab environment or simulator, routers, switches, wireless concepts and network analysis tools.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-031 | CCNA | NT-C-031-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Models, Ethernet, cabling and device roles; IPv4/IPv6, subnetting and address plans
Completion evidence	Topology lab; Subnet workbook
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Network foundation

Objective: Models, Ethernet, cabling and device roles

Observable evidence: Topology lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Models	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when models is appropriate; apply it in a CCNA task; verify the result.	Topology lab
Ethernet	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ethernet is appropriate; apply it in a CCNA task; verify the result.	Topology lab
cabling	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cabling is appropriate; apply it in a CCNA task; verify the result.	Topology lab
device roles	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when device roles is appropriate; apply it in a CCNA task; verify the result.	Topology lab

Module 2 - IP addressing

Objective: IPv4/IPv6, subnetting and address plans

Observable evidence: Subnet workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
IPv4/IPv6	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ipv4/ipv6 is appropriate; apply it in a CCNA task; verify the result.	Subnet workbook
subnetting	A logical subdivision of an ip network. In CCNA, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when subnetting is appropriate; apply it in a CCNA task; verify the result.	Subnet workbook
address plans	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when address plans is appropriate; apply it in a CCNA task; verify the result.	Subnet workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Topology lab; Subnet workbook, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-031 | CCNA | NT-C-031-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	MAC learning, VLANs, trunks and STP; Static routing and single-area OSPF basics
Completion evidence	Switched network; Routed network
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Switching

Objective: MAC learning, VLANs, trunks and STP

Observable evidence: Switched network

Concept	Approved knowledge statement	Application behaviour	Evidence
MAC learning	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when mac learning is appropriate; apply it in a CCNA task; verify the result.	Switched network
VLANs	A logical layer 2 broadcast domain configured on switching infrastructure. In CCNA, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when vlans is appropriate; apply it in a CCNA task; verify the result.	Switched network
trunks	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when trunks is appropriate; apply it in a CCNA task; verify the result.	Switched network
STP	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when stp is appropriate; apply it in a CCNA task; verify the result.	Switched network

Module 4 - Routing

Objective: Static routing and single-area OSPF basics

Observable evidence: Routed network

Concept	Approved knowledge statement	Application behaviour	Evidence
Static routing	Selection and forwarding of network paths between destinations. In CCNA, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when static routing is appropriate; apply it in a CCNA task; verify the result.	Routed network
single-area OSPF basics	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when single-area ospf basics is appropriate; apply it in a CCNA task; verify the result.	Routed network

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Switched network; Routed network, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-031 | CCNA | NT-C-031-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	DHCP, DNS, NTP, NAT and QoS concepts; WLAN components, configuration and security
Completion evidence	Services lab; Wireless design
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - IP services

Objective: DHCP, DNS, NTP, NAT and QoS concepts

Observable evidence: Services lab

Concept	Approved knowledge statement	Application behaviour	Evidence
DHCP	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dhcp is appropriate; apply it in a CCNA task; verify the result.	Services lab
DNS	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dns is appropriate; apply it in a CCNA task; verify the result.	Services lab
NTP	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ntp is appropriate; apply it in a CCNA task; verify the result.	Services lab
NAT	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when nat is appropriate; apply it in a CCNA task; verify the result.	Services lab
QoS concepts	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when qos concepts is appropriate; apply it in a CCNA task; verify the result.	Services lab

Module 6 - Wireless

Objective: WLAN components, configuration and security

Observable evidence: Wireless design

Concept	Approved knowledge statement	Application behaviour	Evidence
WLAN components	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when wlan components is appropriate; apply it in a CCNA task; verify the result.	Wireless design
configuration	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when configuration is appropriate; apply it in a CCNA task; verify the result.	Wireless design
security	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when security is appropriate; apply it in a CCNA task; verify the result.	Wireless design

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Services lab; Wireless design, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-031 | CCNA | NT-C-031-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Device hardening, ACLs, management and API concepts; Verification commands, fault isolation and documentation
Completion evidence	Secured lab; Integrated network
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Security and automation

Objective: Device hardening, ACLs, management and API concepts

Observable evidence: Secured lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Device hardening	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when device hardening is appropriate; apply it in a CCNA task; verify the result.	Secured lab
ACLs	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when acls is appropriate; apply it in a CCNA task; verify the result.	Secured lab
management	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when management is appropriate; apply it in a CCNA task; verify the result.	Secured lab
API concepts	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when api concepts is appropriate; apply it in a CCNA task; verify the result.	Secured lab

Module 8 - Troubleshooting capstone

Objective: Verification commands, fault isolation and documentation

Observable evidence: Integrated network

Concept	Approved knowledge statement	Application behaviour	Evidence
Verification commands	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when verification commands is appropriate; apply it in a CCNA task; verify the result.	Integrated network
fault isolation	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when fault isolation is appropriate; apply it in a CCNA task; verify the result.	Integrated network
documentation	A defined competency within CCNA. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a CCNA task; verify the result.	Integrated network

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Secured lab; Integrated network, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-031 | CCNA | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-031-SK01	Network foundation	Prerequisite foundation	Topology lab	Critical
NT-C-031-SK02	IP addressing	M01 plus earlier foundations	Subnet workbook	Critical
NT-C-031-SK03	Switching	M02 plus earlier foundations	Switched network	Supporting
NT-C-031-SK04	Routing	M03 plus earlier foundations	Routed network	Supporting
NT-C-031-SK05	IP services	M04 plus earlier foundations	Services lab	Supporting
NT-C-031-SK06	Wireless	M05 plus earlier foundations	Wireless design	Critical
NT-C-031-SK07	Security and automation	M06 plus earlier foundations	Secured lab	Supporting
NT-C-031-SK08	Troubleshooting capstone	M07 plus earlier foundations	Integrated network	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-031 | CCNA | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-031-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Network foundation and IP addressing?	The learner should provide Topology lab; Subnet workbook and explain how the work applies Models, Ethernet, cabling and device roles; IPv4/IPv6, subnetting and address plans. Recall alone is insufficient.
NT-C-031-S01-EX02	Scenario/remediation	A learner reports completing IP addressing but cannot produce Subnet workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-031-S01, request Subnet workbook, and allow a new equivalent test only after remediation.
NT-C-031-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Switching and Routing?	The learner should provide Switched network; Routed network and explain how the work applies MAC learning, VLANs, trunks and STP; Static routing and single-area OSPF basics. Recall alone is insufficient.
NT-C-031-S02-EX02	Scenario/remediation	A learner reports completing Routing but cannot produce Routed network. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-031-S02, request Routed network, and allow a new equivalent test only after remediation.
NT-C-031-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of IP services and Wireless?	The learner should provide Services lab; Wireless design and explain how the work applies DHCP, DNS, NTP, NAT and QoS concepts; WLAN components, configuration and security. Recall alone is insufficient.
NT-C-031-S03-EX02	Scenario/remediation	A learner reports completing Wireless but cannot produce Wireless design. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-031-S03, request Wireless design, and allow a new equivalent test only after remediation.
NT-C-031-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Security and automation and Troubleshooting capstone?	The learner should provide Secured lab; Integrated network and explain how the work applies Device hardening, ACLs, management and API concepts; Verification commands, fault isolation and documentation. Recall alone is insufficient.
NT-C-031-S04-EX02	Scenario/remediation	A learner reports completing Troubleshooting capstone but cannot produce Integrated network. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-031-S04, request Integrated network, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-031 | CCNA | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	VLAN campus lab	Switching and addressing	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Routed branch network	OSPF, services and ACLs	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Troubleshooting challenge	Integrated verification and documentation	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-031 | CCNA | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Network Support Engineer Trainee	IP 34% [critical]; switching 33% [critical]; routing 33%	Configured topology	If a critical skill is below 60 or configured topology is missing, return Improve First or Needs Evidence with the linked module/project.
2	NOC Technician - Junior	Monitoring 50% [critical]; fault isolation 50% [critical]	Troubleshooting log	If a critical skill is below 60 or troubleshooting log is missing, return Improve First or Needs Evidence with the linked module/project.
3	Junior Network Administrator	Services 34% [critical]; access 33% [critical]; documentation 33%	Network capstone	If a critical skill is below 60 or network capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-031 | CCNA | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is CCNA for?	Learners preparing for junior networking and NOC support roles.
What should I know before starting?	Basic computer and networking awareness.
How long is the approved learning journey?	12-16 weeks / 140-180 guided hours.
What will I be able to demonstrate?	Configure and troubleshoot routed and switched networks using secure foundational practices.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
CCNA knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-031; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```